Walchand College of Engineering, Sangli

Computer Science & Engineering

Third Year

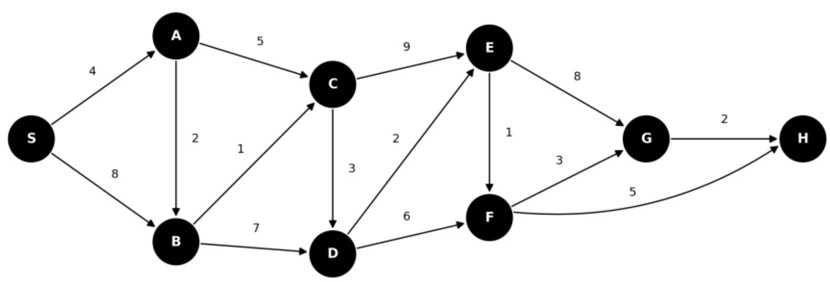
Course: Computer Algorithm Lab

Assignment No. 5

Greedy Method

1. Network Signal Propagation – Shortest Delay Path

A telecom company models its relay network as a directed graph, where each edge represents a one-way transmission link and its weight is the propagation delay (in milliseconds). A signal is broadcast from server node S. Implement Dijkstra's algorithm to compute the minimum time it takes for the signal to reach every other node in the network.



Q) Using the above network with source S, what is the shortest delay to reach node H, and which nodes does that path pass through?

Q) Dijkstra's algorithm assumes all edge weights are non-negative. If one link in this network had a negative propagation delay, would the algorithm still return correct shortest delays for every node? Justify your answer.

Q) This network happens to be a Directed Acyclic Graph (DAG). Explain how the shortest paths from S could instead be computed using topological sorting, and compare the time complexity of that approach with Dijkstra's algorithm.

Q) If a new, faster link is added between two nodes that already had a connecting path, does the entire shortest path tree need to be recomputed from scratch? Explain your reasoning.

2. Esports Arena – Match Scheduling Problem

A gaming arena has a single main stage and receives 25 requests to host matches throughout a tournament day, each with a start time and an end time. Using a greedy approach, select the maximum number of non-overlapping matches that can be hosted on the main stage.

Input sample:

Matches (start, end) = [(1,3), (2,5), (0,6), (5,8), (4,9), (6,10), (8,11), (9,13), (11,14), (2,15), (13,17), (12,16), (15,18), (17,19), ...]

Implement the Activity Selection problem for this scenario.

3. Corporate Office – Minimum Meeting Rooms Required

A corporate office receives 25 meeting requests for a single day, each with a start time and an end time. Determine the minimum number of meeting rooms required so that no two meetings scheduled in the same room overlap.

Input sample:

Meetings (start, end) = [(900,1000), (930,1330), (1200,1300), (1330,1500), (1400,1430), (1500,1700), (1600,1630), ...]

Implement the Minimum Meeting Rooms problem.

4. Shared Office Printer – Minimizing Total Waiting Time

A shared office printer has a queue of 40 pending print jobs submitted by different employees, each taking a different amount of time to complete. Using a greedy strategy, decide the order in which jobs should be printed so that the total (and average) waiting time across all employees is minimized.

Input sample:

Print job durations (seconds) = [45, 120, 30, 80, 200, 10, 150, 70, 250, 40, 180, 90, 20, 300, 60, ...]

Implement an algorithm to minimize the total waiting time, and report the resulting order and the total/average waiting time.

5. Marketing Campaign – Minimum Channel Coverage

A marketing firm wants to run a campaign that reaches 15 target demographic segments (D1–D15). Each advertising channel (TV, radio, social media platform, etc.) reaches a fixed subset of these segments, and running an ad on a channel has a cost, so the firm wants to use as few channels as possible. Using a greedy approach, select the minimum number of channels such that every demographic segment is reached by at least one selected channel.

Input sample:

Channel1 reaches {D1, D2, D5, D9}

Channel2 reaches {D2, D3, D4}

Channel3 reaches {D4, D5, D6, D7}

Channel4 reaches {D8, D9, D10, D11}

Channel5 reaches {D10, D12, D13, D14, D15}

Channel6 reaches {D1, D6, D7, D8}

Channel7 reaches {D3, D11, D12}

Channel8 reaches {D13, D14, D15} ... (10 candidate channels in total)

Implement the greedy Set Cover approximation algorithm.

6. ATM Cash Dispenser – Minimum Notes Problem

An ATM must dispense a requested withdrawal amount to a customer using the minimum possible number of currency notes, chosen from the denominations it stocks.

Input sample:

Denominations available = [1, 2, 5, 10, 20, 50, 100, 200, 500]

Amount to be withdrawn = 1287

Implement the greedy Minimum Notes (Coin Change) algorithm. In addition, construct one set of denominations for which the greedy approach does NOT give the optimal answer, and justify why it fails.